

ANDY YANG

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TECHNICAL SKILLS

Languages: C++, Python, JavaScript, Typescript, SQL, HTML, CSS, C
Frameworks/Libraries: React, React Redux, Express.js, Mongoose
Developer Tools: Git, Bazel, Node.js, MongoDB, ClickHouse, MySQL, Protocol Buffers

EDUCATION

University of Waterloo Waterloo, ON
Bachelor of Computer Science September 2022 - April 2027 (Expected)

EXPERIENCE

Software Engineer May 2026 - Present
BitGo

- Building and optimizing **full-stack features** on the Web Experience Team to support secure, high-performance crypto finance platforms

Backend Engineer April 2025 - November 2025
Tencent - WeChat Search Team, Beijing Headquarters

- Designed and implemented a **Last Finishing Child based critical path algorithm** in C++ to pinpoint bottlenecks in **RPC traces**, enabling **targeted debugging and optimization**
- Engineered a **pipelined trace-aggregation system** to ingest millions of daily traces in **chunked streams**, constructing dependency graphs to calculate critical paths within **strict container memory limits**.
- Implemented **log correlation logic** in the logging pipeline to collect logs by request ID, match caller-callee RPC log pairs, and calculate and report request-level metrics, enabling **real-time performance monitoring per module**

Software Engineer January 2025 - April 2025
Big Data Center

- Engineered a **RAG chat pipeline** in Python with **web retrieval** and **chat history** for accurate, context-aware responses
- Developed an **adaptive chunking algorithm** to segment text into large and small chunks with context overlap, allowing more targeted retrieval without overall context loss
- Combined **BM25** and **vector-search** results with **RRF** for **RAG retrieval** in a **small-to-big** pipeline to improve search accuracy and context retention
- Built a **concurrent Python scraping engine** using **thread pools** to extract real-time web search payloads with **low latency** for live RAG augmentation

Full-Stack Developer April 2024 - August 2024
Chase Auto Parts

- Built a full-stack search engine web application, providing **low latency** search for **thousands** of parts and items
- Implemented **keyword search** and filtering via **dynamic SQL queries** in tandem with **MS-SQL's full-text search**
- Reduced upload and update latency** for larger datasets by leveraging **WebSocket streams**, enabling **continuous data flow** without repeated HTTP overhead and enhancing user experience through **real-time status updates**

PROJECTS

🔗 **ASL Translator** | Python, PyTorch, OpenCV, MediaPipe, CUDA April 2026 - Present

- Designed a **Landmark-to-Sequence** architecture isolating skeletal features from raw video, **reducing input dimensionality by >99%** to enable **real-time edge execution**
- Engineered a **Stacked LSTM network** in PyTorch for spatial-temporal modeling, prioritizing a **minimal VRAM footprint** over Transformer architectures to ensure real-time execution on local hardware
- Integrating **CTC loss** alongside a fine-tuned local **T5-Small Transformer** to decode unsegmented gestures and perform offline, real-time ASL-to-English syntax correction

Extracurriculars: UW Computer Science Club, UW Data Science Club, UW Game Dev Club, UWCSA
Hobbies/Interests: Piano, Violin, Guitar, Badminton, Swimming, Video Game Design, Computer Hardware